



CYBERSECURITY DEPARTMENT • INTERNSHIP PROJECT

Intrusion Prevention System with SNORT + Alert Dashboard

Detecting and blocking network attacks in a self-built lab environment

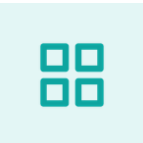
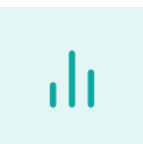
Presented to: Supervisor & IT Department

Rezzag Nadjia

Guellil Imane

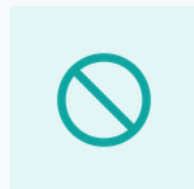
hachani chaima

What We'll Cover

01		Objectives	What we set out to build and why
02		Technologies Used	Tools and stack behind the project
03		System Architecture	How the network and devices are laid out
04		How the System Works	Traffic flow and detection logic, step by step
05		Results Achieved	Attack testing outcomes and the live dashboard

Objectives

Deploy a working Intrusion Prevention System using SNORT that detects and actively blocks common network attacks — with a live dashboard so alerts don't have to be read from raw logs.



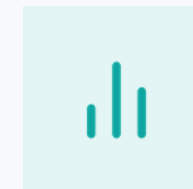
Detect & Block

Identify malicious traffic (SQLi, XSS, traversal, scans) and drop it in real time



Realistic Lab

Client / SNORT / Server across two isolated networks, SNORT sitting inline



Visibility

Dashboard surfacing alerts, sources, and attack types as they happen

Technologies Used

Everything the lab and dashboard are built on



Ubuntu Server

Host OS for all three VMs — CLIENT, SNORT, and SERVER



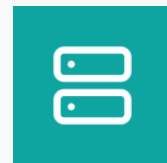
VirtualBox

Virtualization platform hosting the isolated lab network



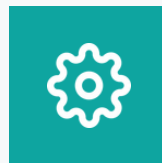
SNORT (IDS/IPS)

Core detection & prevention engine — rule-based packet inspection



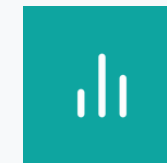
Apache HTTP Server

Target web service on SERVER, used to simulate real traffic



Netplan / IP Forwarding

Static network configuration and routing between segments

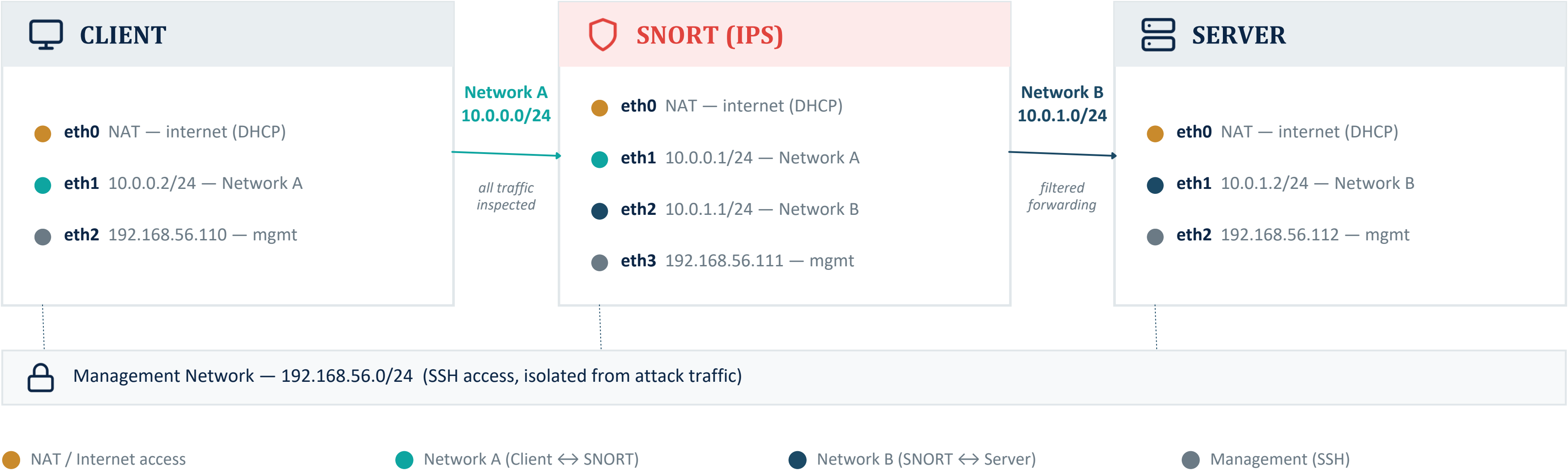


Dashboard Stack

use python flask framework

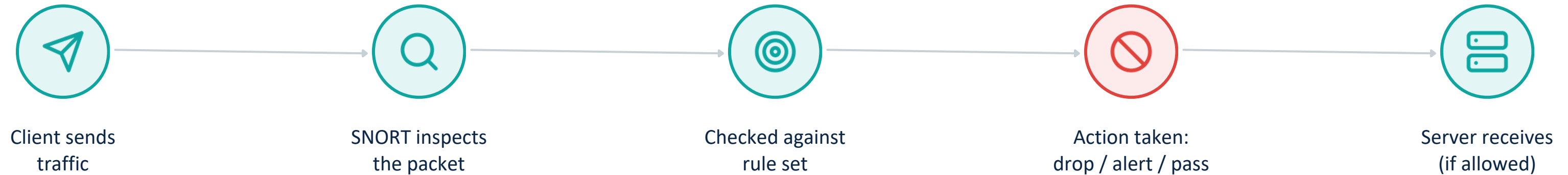
System Architecture

SNORT sits inline between two isolated networks, plus separate NAT and management paths



How the System Works

Every packet between CLIENT and SERVER is inspected by SNORT before it's allowed through



DROP

Blocks the packet and logs it. Used for confirmed attacks (SQLi, traversal, XSS).



ALERT

Logs the event but lets it through. Used to monitor suspicious-but-unconfirmed traffic.



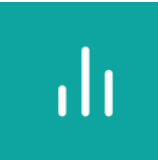
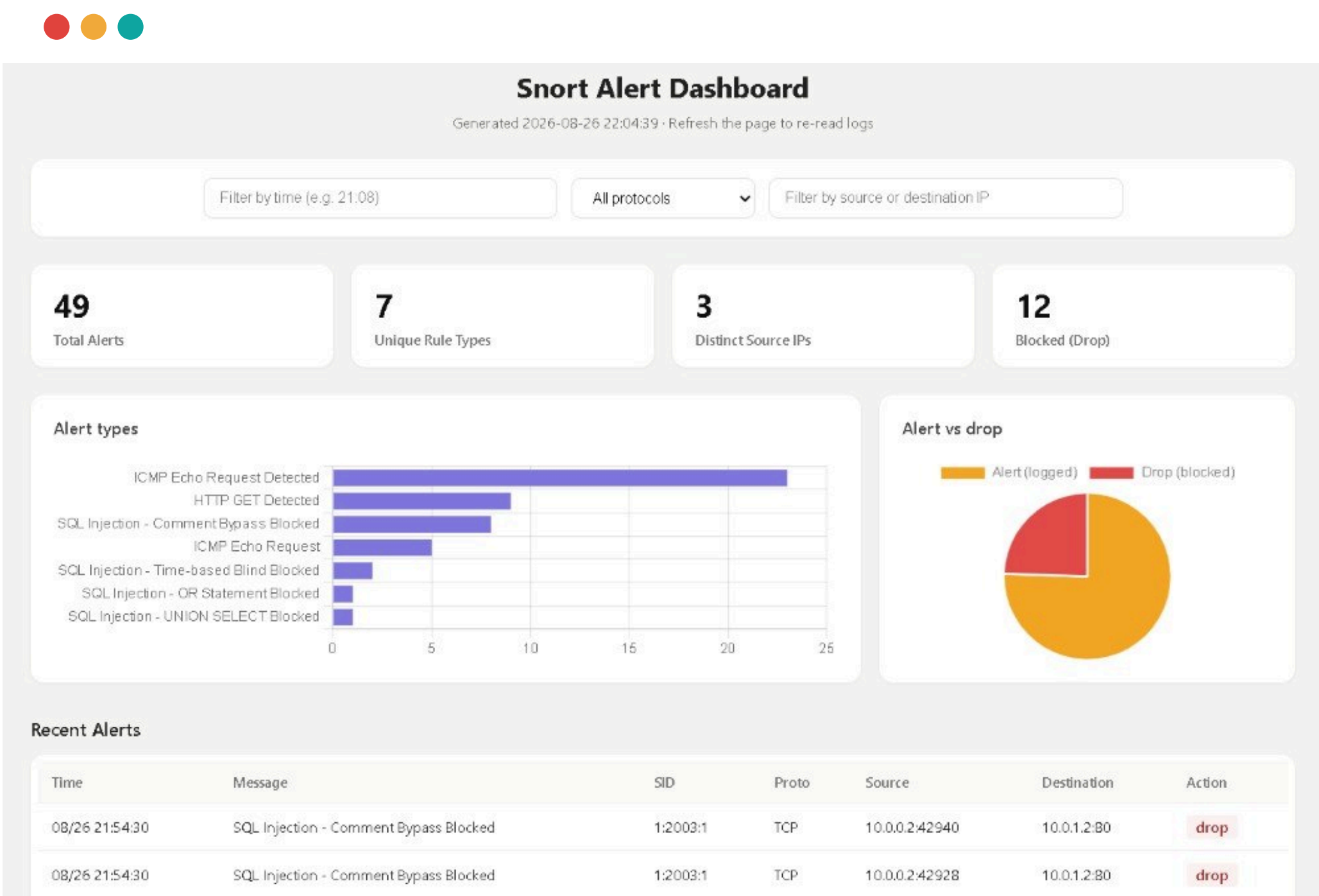
PASS

Allows silently, no log. Used to whitelist trusted internal or partner traffic.

```
drop tcp any any -> any 80 (msg:"SQL Injection - UNION SELECT Blocked"; content:"UNION"; http_uri;  
content:"SELECT"; http_uri; sid:2001;)
```

Live Alert Dashboard

Turns raw SNORT log lines into something the team can actually monitor



Live alert count by severity and rule type



Source IP, destination, and timestamp per event



Breakdown of attack categories (SQLi, XSS, scans...)



Replaces manually reading `/var/log/snort/alert`



Thank you

happy to walk through a live demo of the lab and dashboard.